

Materials, Supply Chain & Price Index Report

International material indices, supplier concentration, logistics, lead times, and procurement exposure.

Project: Solar Alpha **Location:** Solano County, California (fictional demo site) **Date:** August 14, 2026

Demo note: This document contains intentionally synthetic data and seeded inconsistencies for a one-day hackathon prototype. It is not a real engineering, legal, environmental, investment, or regulatory opinion.

Procurement summary

The project cost plan was prepared using Q1 2026 budget quotes. Since then, several high-value components have moved materially in price and lead time.

Material / package	Base index (Jan=100)	Aug 2026 index	Change	Lead time
Power transformers	100	128	+28%	62 weeks
Steel structures	100	117	+17%	28 weeks
Copper cable	100	122	+22%	24 weeks
Battery containers	100	109	+9%	34 weeks
PV modules	100	94	-6%	18 weeks

Indices are synthetic Project Red Flag benchmark series representing an international landed-cost composite for demo purposes.

Transformer exposure

The electrical design requires two main power transformers. The procurement plan names one technically qualified supplier. The schedule allows 38 weeks from purchase order to site delivery, while the latest supplier indication states 62 weeks.

CRITICAL RED FLAG - Transformer lead time exceeds schedule allowance by 24 weeks and there is no validated alternate supplier.

Cost-plan reconciliation

Package	Q1 budget (\$M)	Updated estimate (\$M)	Variance
Transformers	14.2	18.6	+4.4
Steel / trackers	23.5	27.1	+3.6
Copper / electrical BOS	16.0	19.1	+3.1
Battery system	46.0	50.1	+4.1
PV modules	42.5	40.0	-2.5

Net indicated increase across the sampled packages is approximately \$12.7M before contingency effects. The current finance memo still references total project CAPEX of \$186M.

RED FLAG - Updated package pricing suggests total CAPEX could approach \$199-211M depending on contingency and balance-of-plant pass-throughs.

Logistics

Oversize transformer deliveries are assumed to use the private heavy-haul easement referenced in the land report. If unavailable, rerouting adds distance and may require bridge-load verification.

Project Red Flag extraction targets

Material index change %, indexed procurement exposure, quote age, lead-time variance, supplier concentration, logistics dependency, currency exposure, and contingency adequacy.